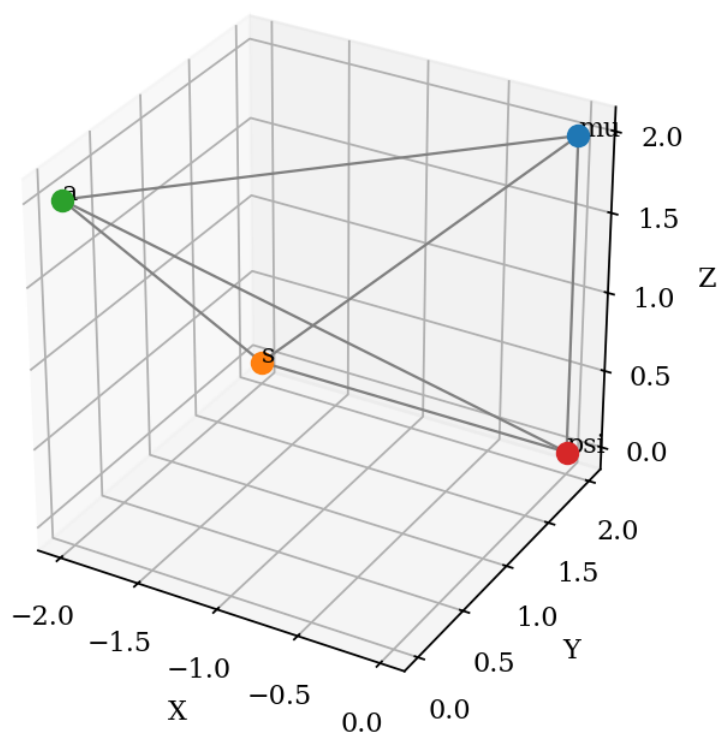


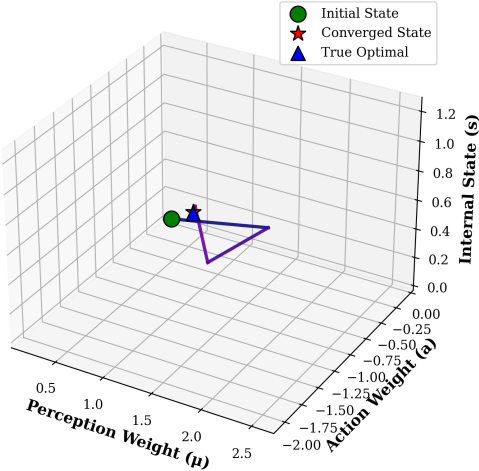
$$\begin{matrix} \mu \\ s \\ a \\ \psi \end{matrix}$$

$$\begin{matrix} \mu s \nabla_{\mu} F \\ a \psi \nabla_a F \end{matrix}$$

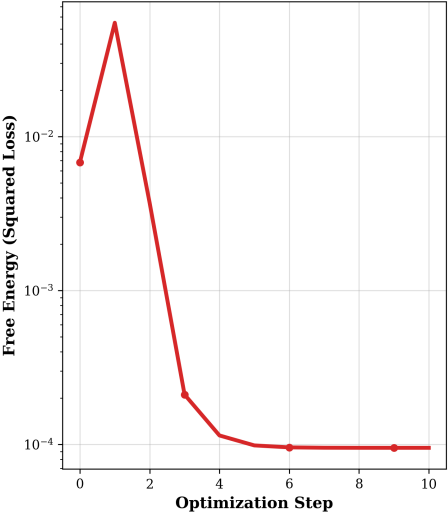
Four-fold partition mapped to Quadray tetrahedron



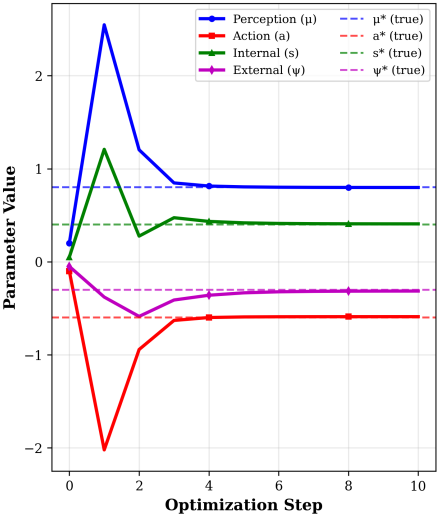
4D Natural Gradient Trajectory
(Active Inference Dynamics)



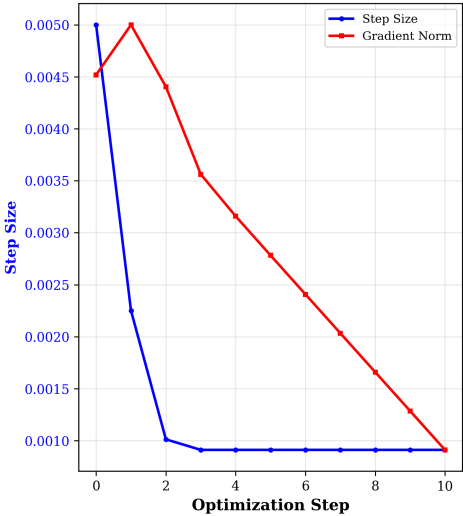
Free Energy Minimization
(Active Inference Principle)



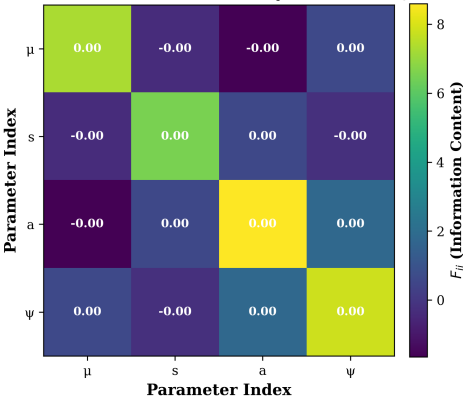
Four-Fold Partition Evolution
(Fuller.4D Tetrahedral Structure)



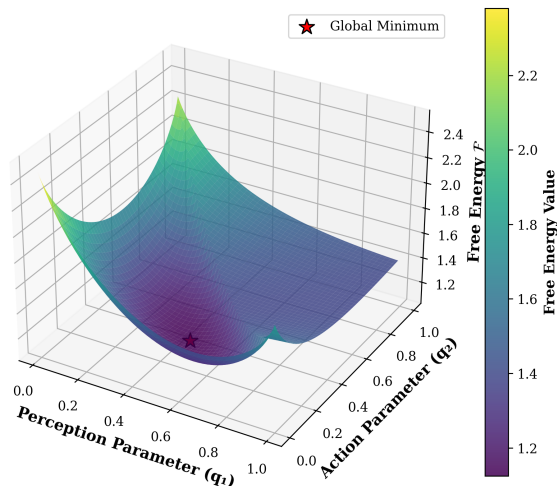
Optimization Diagnostics
(Step Size & Gradient Evolution)



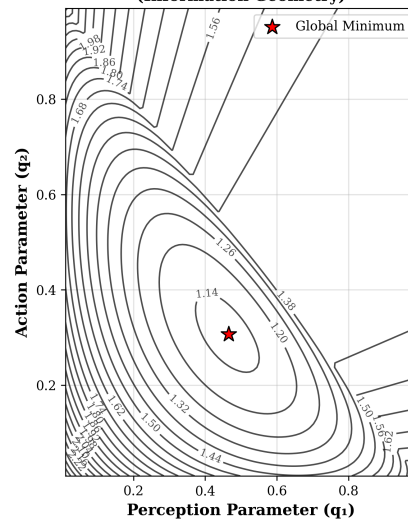
Fisher Information Matrix
(Information Geometry Metric)



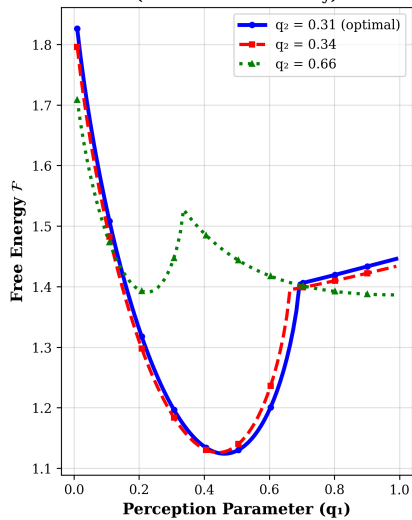
Free Energy Landscape
(4D Active Inference Framework)



Free Energy Contours
(Information Geometry)



Free Energy Cross-Sections
(Parameter Sensitivity)



Active Inference Four-Fold Partition:

μ (Internal States)
 s (Sensory) a (Actions)
 ψ (External Causes)

4D Framework Integration:

- Fuller.4D: Tetrahedral structure with IVM quantization
- Coxeter.4D: Exact Euclidean geometry for measurements
- Einstein.4D: Information-geometric flows and geodesics

Free Energy Principle:

- Minimization drives perception-action cycles
- Balances prediction error vs. complexity
- Natural gradient follows geodesics on information manifold

Mathematical Structure:

- Variational posterior $Q(s)$ over latent states
- Prior $P(s)$ encoding environmental structure
- Likelihood $P(o|s)$ connecting observations to states

Local Curvature
(Fisher Information Structure)

